

- 1 (i) For $\epsilon > 0$, $P(|X_n - X| > \epsilon) = P(X_n = n) = \frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$. Hence $X_n \rightarrow X$ in probability.
- (ii) For $x = 1$, $X_n(1) = X(1) = 0$. So $X_n(1) \rightarrow X(1)$ as $n \rightarrow \infty$. For $x = 0$, $X(0) = 0$ and $X_n(0) = n$, $\forall n$. So $X_n(0) \not\rightarrow X(0)$ as $n \rightarrow \infty$. For $x \in (0, 1)$, then $\exists n_0$ s.t. $\frac{1}{n_0} < x$ and $\forall n \geq n_0$, $X_n(x) = 0 = X(x)$. So $X_n(x) \rightarrow X(x)$, $x \in (0, 1)$ as $n \rightarrow \infty$. So $\{x : X_n(x) \not\rightarrow X(x)\} = \{0\}$ and $P(\{0\}) = 0$. Hence $X_n \rightarrow X$ a.s. (P).
- (iii) Since $\int |X_n| I_{\{|X_n| > T\}} dP = nm([0, \frac{1}{n})) = 1$, $\forall n > T$, then $\sup_{n \geq 1} \int |X_n| I_{\{|X_n| > T\}} dP \geq 1$. So $\lim_{T \rightarrow \infty} \sup_{n \geq 1} \int |X_n| I_{\{|X_n| > T\}} dP \neq 0$. Hence X_n is not UI.
- (iv) $\int |X_n - X|^p dP = \int n^p I_{[0, \frac{1}{n})} dm = n^p \cdot \frac{1}{n} = n^{p-1} \rightarrow 0$. So $X_n \rightarrow X$ in L^p for $0 < p < 1$.
2. Here $P_{X_n}(\{n\}) = \frac{1}{n} = 1 - P_{X_n}(\{0\})$. Note that this is same as in Problem 1.
- (i) For $\epsilon > 0$, $P(|X_n - X| > \epsilon) = P(X_n > \epsilon) = P(X_n = n) = P_{X_n}(\{n\}) = \frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$. Hence $X_n \rightarrow X$ in probability.
- (ii) Without knowing the Ω, X_n, X , we can not say for sure (using results we know so far) if $X_n \rightarrow X$ a.s. (P). For example, we can not produce a null set needed for almost sure convergence.
- (iii) Using change of variable $\int |X_n| I_{\{|X_n| > T\}} dP = \int |x| I_{\{|x| > T\}} dP_{X_n}(x) = nm([0, \frac{1}{n})) = 1$, $\forall n > T$. So, $\sup_{n \geq 1} \int |X_n| I_{\{|X_n| > T\}} dP \geq 1$ and $\lim_{T \rightarrow \infty} \sup_{n \geq 1} \int |X_n| I_{\{|X_n| > T\}} dP \neq 0$. Hence X_n is not UI.
- (iv) $\int |X_n - X|^p dP = \int |X_n|^p dP = \int |x|^p dP_{X_n}(x) = n^p \cdot \frac{1}{n} = n^{p-1} \rightarrow 0$. So $X_n \rightarrow X$ in L^p for $0 < p < 1$.
3. (a) Let $g(x) = x^{\frac{p}{q}}, x > 0$ and $\frac{p}{q} > 1$, then $g(x)$ is a convex function. Since $f_n \rightarrow f$ in L^p , then $(\int |f_n - f|^p d\mu) \rightarrow 0$ as $n \rightarrow \infty$. By Jensen's inequality, we have $g(\int |f_n - f|^q d\mu) \leq \int g(|f_n - f|^q) d\mu$, i.e. $(\int |f_n - f|^q d\mu)^{\frac{p}{q}} \leq \int |f_n - f|^p d\mu$. Then $\int |f_n - f|^q d\mu \leq (\int |f_n - f|^p d\mu)^{\frac{q}{p}} \rightarrow 0$, as $n \rightarrow \infty$, i.e. $(\int |f_n - f|^q d\mu)^{\frac{1}{q}} \rightarrow 0$ as $n \rightarrow \infty$. Hence $f_n \rightarrow f$ in L^q for any $0 < q < p$.
- (b) **DCT:** Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f_n, f, g be measurable functions. If $|f_n| \leq g$, a.e. (μ), $\forall n \geq 1$, $\int g d\mu < \infty$ and $f_n \rightarrow f$ a.e. (μ), then $f \in L^1(\Omega, \mathcal{F}, \mu)$, $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$ and $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.
- To prove the second part (when $f_n \rightarrow f$ in measure (μ)), we will use this real analysis fact: A sequence $\{x_n\}$ converges to x iff for any subsequence $\{x_{n_k}\}$ of $\{x_n\}$, there is a further subsequence $\{x_{n_{k_l}}\}$ of $\{x_{n_k}\}$ such that $x_{n_{k_l}} \rightarrow x$. Denote $a_n = \int f_n d\mu$, $a = \int f d\mu$ and $b_n = \int |f_n - f| d\mu$. Will show $a_n \rightarrow a$ and $b_n \rightarrow 0$. Choose any subsequence $\{a_{n_k}\}$ of $\{a_n\}$, and for this subsequence, choose the corresponding subsequence functions $\{f_{n_k}\}$ of $\{f_n\}$. Since $f_n \rightarrow f$ in measure, then $\lim_{n \rightarrow \infty} \mu(|f_n - f| > \epsilon) = 0$, $\forall \epsilon > 0$, thus $\lim_{k \rightarrow \infty} \mu(|f_{n_k} - f| > \epsilon) = 0$, $\forall \epsilon > 0$, i.e. $f_{n_k} \rightarrow f$ in measure. By Thm 2.5.2, we can get a sub-subsequence $f_{n_{k_l}} \rightarrow f$ a.e. (μ). For this sub-subsequence $\{f_{n_{k_l}}\}$, conditions of DCT are true. Hence from DCT, we have $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\lim_{l \rightarrow \infty} \int f_{n_{k_l}} d\mu = \int f d\mu$, i.e. $a_{n_{k_l}} \rightarrow a$. By Fact 1, we get $a_n \rightarrow a$, i.e. $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$. Similarly, we have $\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.
- (c) With $g_n = g - f_n \geq 0$, apply Fatou's Lemma to get $\liminf_{n \rightarrow \infty} \int g_n d\mu \geq \int (\liminf_{n \rightarrow \infty} g_n) d\mu$. Hence, $\int f d\mu - \limsup_{n \rightarrow \infty} \int f_n d\mu \geq \int f d\mu - \int (\limsup_{n \rightarrow \infty} f_n) d\mu$. Now using the fact that $f \in L^1$, we get the result.